

Series Review 2 (covers 5.2 – 5.6)

Directions: Determine if each of the following series converges or diverges. Use any method.

1. $\sum_{k=1}^{\infty} \left(\frac{k+2}{3k-1} \right)^k$

$$\rho = \lim_{k \rightarrow \infty} \sqrt[k]{\left(\frac{k+2}{3k-1} \right)^k} = \lim_{k \rightarrow \infty} \frac{k+2}{3k-1} = \frac{1}{3} < 1. \text{ Therefore, by the root test, the series } \sum_{k=1}^{\infty} \left(\frac{k+2}{3k-1} \right)^k \text{ converges.}$$

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2. $\sum_{k=1}^{\infty} \frac{3^k k^2}{k!}$

$$\rho = \lim_{k \rightarrow \infty} \frac{\left| \frac{3^{k+1} (k+1)^2}{(k+1)!} \right|}{\left| \frac{3^k k^2}{k!} \right|} = \lim_{k \rightarrow \infty} \frac{3^{k+1} (k+1)^2}{(k+1)!} \cdot \frac{k!}{3^k k^2} = \lim_{k \rightarrow \infty} \frac{3(k+1)^2}{(k+1)k^2} = \lim_{k \rightarrow \infty} \frac{3(k+1)}{k^2} = 0 < 1. \text{ Therefore, by the ratio}$$

test, the series $\sum_{k=1}^{\infty} \frac{3^k k^2}{k!}$ converges.

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3. $\sum_{k=1}^{\infty} [\tan^{-1}(k+1) - \tan^{-1}(k)]$

This appears to be a telescoping series. We will try to find a formula for the partial sum s_n .

$$s_n = (\tan^{-1}(2) - \tan^{-1}(1)) + (\tan^{-1}(3) - \tan^{-1}(2)) + \dots + (\tan^{-1}(n+1) - \tan^{-1}(n)) = (\tan^{-1}(n+1) - \tan^{-1}(1))$$

We note that $\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} (\tan^{-1}(n+1) - \tan^{-1}(1)) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$. Therefore, since the sequence of partial

sums converges to $\frac{\pi}{4}$, $\sum_{k=1}^{\infty} [\tan^{-1}(k+1) - \tan^{-1}(k)] = \frac{\pi}{4}$.

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$$4. \sum_{k=1}^{\infty} \frac{k^{4/3}}{8k^2 + 5k + 1}$$

We note that $\sum_{k=1}^{\infty} \frac{k^{4/3}}{8k^2 + 5k + 1}$ is likely to behave like $\sum_{k=1}^{\infty} \frac{k^{4/3}}{8k^2} = \sum_{k=1}^{\infty} \frac{1}{8k^{2/3}}$. We then investigate:

$$\lim_{k \rightarrow \infty} \frac{\left(\frac{k^{4/3}}{8k^2 + 5k + 1} \right)}{\left(\frac{1}{8k^{2/3}} \right)} = \lim_{k \rightarrow \infty} \frac{k^{4/3}}{8k^2 + 5k + 1} \cdot \frac{8k^{2/3}}{1} = \lim_{k \rightarrow \infty} \frac{8k^2}{8k^2 + 5k + 1} = 1. \text{ Therefore, by the limit comparison test,}$$

the series $\sum_{k=1}^{\infty} \frac{k^{4/3}}{8k^2 + 5k + 1}$ and $\sum_{k=1}^{\infty} \frac{1}{8k^{2/3}}$ must either both converge or diverge. However, $\sum_{k=1}^{\infty} \frac{1}{8k^{2/3}}$ is a p -series with $p = \frac{2}{3} < 1$ and so it diverges. Therefore, $\sum_{k=1}^{\infty} \frac{k^{4/3}}{8k^2 + 5k + 1}$ diverges. ■

$$5. \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2 + 1}$$

We note that this is an alternating series. If we consider $a_k = \frac{1}{k^2 + 1}$ we note that $a_{k+1} \leq a_k$ and $\lim_{k \rightarrow \infty} a_k = 0$.

Therefore, by the alternating series test, the series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2 + 1}$ is convergent. ■

$$6. \sum_{k=1}^{\infty} \frac{(2)^{k+1}}{2k}$$

Let $f(x) = \frac{(2)^{x+1}}{2x} = \frac{2^x}{x}$. We note that $\lim_{x \rightarrow \infty} \frac{2^x}{x} = \lim_{x \rightarrow \infty} \frac{(2)^x \ln(2)}{1} = \infty$. Therefore, by the test for divergence, the

series $\sum_{k=1}^{\infty} \frac{(2)^{k+1}}{2k}$ is divergent. ■

7.
$$\sum_{k=1}^{\infty} \frac{1}{k(k+2)}$$

Using partial fractions we can rewrite the series as follows: $\sum_{k=1}^{\infty} \frac{1}{k(k+2)} = \sum_{k=1}^{\infty} \left(\frac{1}{2k} - \frac{1}{2(k+2)} \right)$. This series

appears to be telescoping, so we will look at the sequence of partial sums s_n .

$$s_n = \left(\frac{1}{2} - \frac{1}{6} \right) + \left(\frac{1}{4} - \frac{1}{8} \right) + \left(\frac{1}{6} - \frac{1}{10} \right) + \left(\frac{1}{8} - \frac{1}{12} \right) + \left(\frac{1}{10} - \frac{1}{14} \right) + \dots + \left(\frac{1}{2n} - \frac{1}{2n+4} \right) = \frac{1}{2} + \frac{1}{4} - \frac{1}{2n+4} = \frac{3}{4} - \frac{1}{2n+4}.$$

Then we see that $\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(\frac{3}{4} - \frac{1}{2n+4} \right) = \frac{3}{4}$. Therefore $\sum_{k=1}^{\infty} \frac{1}{k(k+2)} = \frac{3}{4}$.

8. Classify each of the following series as absolutely convergent, conditionally convergent, or divergent.

a)
$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{10k}$$

We first check to see if the given series is absolutely convergent. We note that

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1}}{10k} \right| = \sum_{k=1}^{\infty} \frac{1}{10k} = \frac{1}{10} \sum_{k=1}^{\infty} \frac{1}{k} \text{ which is the harmonic series and therefore diverges. So the series is not}$$

absolutely convergent.

We will now use the alternating series test to see if the given series converges conditionally.

We let $b_k = \frac{1}{10k}$. We see that $b_{k+1} < b_k$ for all k . Also, $\lim_{k \rightarrow \infty} b_k = 0$. Therefore by the alternating series test

the series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{10k}$ converges.

So the given series is conditionally convergent.



b) $\sum_{k=1}^{\infty} \frac{\sin(k)}{k^3}$

We first check to see if the given series is absolutely convergent. We note that

$$\sum_{k=1}^{\infty} \left| \frac{\sin(k)}{k^3} \right| = \sum_{k=1}^{\infty} \frac{|\sin(k)|}{k^3} \leq \sum_{k=1}^{\infty} \frac{1}{k^3} \text{ which is a } p\text{-series with } p=3 > 1 \text{ and it therefore convergent. Therefore,}$$

by the comparison theorem the series $\sum_{k=1}^{\infty} \left| \frac{\sin(k)}{k^3} \right|$ is convergent. So, $\sum_{k=1}^{\infty} \frac{\sin(k)}{k^3}$ is absolutely convergent. ■

c) $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} 3^{2k-1}}{k^2 + 1}$

We first check to see if the given series is absolutely convergent. We note that $\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1} 3^{2k-1}}{k^2 + 1} \right| = \sum_{k=1}^{\infty} \frac{3^{2k-1}}{k^2 + 1}$.

Now, using the ratio test we see that:

$$\rho = \lim_{k \rightarrow \infty} \frac{3^{2k+1}}{\frac{(k+1)^2 + 1}{3^{2k-1}}} = \lim_{k \rightarrow \infty} \frac{3^{2k+1}}{(k+1)^2 + 1} \cdot \frac{k^2 + 1}{3^{2k-1}} = \lim_{k \rightarrow \infty} \frac{9(k^2 + 1)}{(k+1)^2 + 1} = \lim_{k \rightarrow \infty} \frac{9k^2 + 9}{k^2 + 2k + 2} = 9 > 1. \text{ So, by the ratio test}$$

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1} 3^{2k-1}}{k^2 + 1} \text{ is divergent.}$$
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